Image Restoration

1. Reading the Corrupted Image

Why: First, I loaded the corrupted image into MATLAB because every restoration step works on this image. It contains salt-and-pepper noise and periodic grid noise.

How: I used imread() to read the image and imshow() to inspect the corruption before processing.

Code: i = imread('cursed_schematic.png'); figure; imshow(i);

Screenshot: Leave space here to paste the screenshot of the Original Image.

2. Median Filtering

Why: Salt-and-pepper noise affects individual pixels, so it should be removed before applying FFT.

How: A 3×3 median filter replaces each pixel with the median of its neighbours, preserving edges while removing impulse noise.

Code: j = medfilt2(i,[3 3]);

Screenshot: Paste the screenshot of the Median Filter output here.

3. 2D FFT

Why: Periodic noise is easier to identify in the frequency domain.

How: I applied fft2() followed by fftshift() and displayed the log magnitude spectrum.

Code: f = fft2(j); f = fftshift(f);

Screenshot: Paste the FFT spectrum here.

4. Frequency Mask

Why: Bright peaks correspond to periodic interference and should be removed without affecting the centre frequencies.

How: I ignored the centre, detected peaks above 90% of the maximum value, and created a notch mask around each peak.

Code: threshold = 0.9*max(s(:)); [row,col]=find(s>threshold);

Screenshot: Paste the Frequency Mask here.

5. Image Reconstruction

Why: After removing unwanted frequencies, the filtered image must be converted back to the spatial domain.

How: The inverse FFT reconstructs the restored image from the filtered spectrum.

Code: recovered = real(ifft2(ifftshift(f_filtered)));

Screenshot: Paste the Recovered Image here.

Conclusion

The restoration process successfully combines spatial-domain median filtering with frequency-domain notch filtering. The final recovered image clearly reveals the hidden schematic while removing most of the periodic interference.